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Class : 12

Register Number

COMMON QUARTERLY EXAMINATION - 2026 - 27

Time Allowed : 3.00 Hours] **MATHEMATICS** [Max. Marks : 90

PART - I 20x1=20

Note: 1. All the Questions Are Compoulsory. 2. Choose the most appropriate answer four Options

1. If A, B and C are invertible matrices of some order, then which one of the following is not true?

(1) $adj\ A = |A|\ A^{-1}$ (2) $adj(AB) = (adjA)(adj\ B)$

(3) $detA^{-1} = (\det A)^{-1}$ (4) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

2. If $x^a y^b = e^m, x^c y^d = e^n, \Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}, \Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}, \Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then the values of x and y are respectively.....

(1) $e^{(\Delta_2/\Delta_1)}, e^{(\Delta_3/\Delta_1)}$, (2) $\log(\Delta_1/\Delta_3)$, $\log(\Delta_2/\Delta_3)$

(3) $\log(\Delta_2/\Delta_1)$, $\log(\Delta_3/\Delta_1)$ (4) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$

3. If $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$ then the value of $|z_1 + z_2 + z_3|$ is..

(1) 1 (2) 2 (3) 3 (4) 4

4. z_1, z_2 and z_3 are complex number such that $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3| = 1$

Then $z_1^2 + z_2^2 + z_3^2$ is .....

(1) 3 (2) 2 (3) 1 (4) 0

5. If $\omega \neq 1$ is a cubic root of unity and $(1+\omega)^7 = A + B\omega$, then (A, B) equals to .....

(1) (1,0) (2) (−1,1) 3 (0,1) (4) (1,1)

6. If $\omega \neq 1$ is a cube root of unity and $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is .....

(1) 1 (2) -1 (3) $\sqrt{3}\,i$ (4) $-\sqrt{3}\,i$

7. A zero of $x^3 + 64$ is .....

(1) 0 (2) 4 (3) $4i$ (4) -4

8. The number of positive zeros of the polynomial $\sum_{r=0}^{n} {}^nC_r (-1)^r x^r$ is .....

(1) 0 (2) n (3) $< n$ (4) r

9. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}\ \vec{c}\ \vec{b}]$ is equal to

(1) 2 (2) -1 (3) 1 (4) 0

10. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101}+y^{101}+z^{101}}$ is.....

(1) 0 (2) 1 (3) 2 (4) 3

11. If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to .....

(1) $[-1, 1]$ (2) $[\sqrt{2}, 2]$ (3) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$ (4) $[-2, -\sqrt{2}] \cap [\sqrt{2}, 2]$

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12. $sin(\tan^{-1}x)$, $|x|<1$ is equal to.....

(1) $\frac{x}{\sqrt{1-x^2}}$ (2) $\frac{1}{\sqrt{1-x^2}}$ (3) $\frac{1}{\sqrt{1+x^2}}$ (4) $\frac{x}{\sqrt{1+x^2}}$

13. The equation of the circle passing through (1,5) and (4,1) and touching y-axis is $x^2+y^2-5x-6y+9+\lambda(4x+3y-19)=0$ where λ is equal to .....

(1) $0,-\frac{40}{9}$ (2) 0 (3) $\frac{40}{9}$ (4) $-\frac{40}{9}$

14. The equation of the normal to the circle $x^2+y^2-2x-2y+1=0$ which is parallel to the line $2x+4y=3$ is

(1) $x+2y=3$ (2) $x+2y+3=0$ (3) $2x+4y+3=0$ (4) $x-2y+3=0$

15. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ parallel to the straight line $2x-y=1$. One of the points of contact of tangents on the hyperbola is

(1) $\left(\frac{9}{2\sqrt{2}},\frac{-1}{\sqrt{2}}\right)$ (2) $\left(\frac{-9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (3) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (4) $\left(3\sqrt{3},-2\sqrt{2}\right)$

16. If $0\le\theta\le\pi$, the system of equations $x+(sin\theta)y-(cos\theta)z=0$, $(cos\theta)x-y+z=0$, $(sin\theta)x+y-z=0$ has a non-trivial solution then θ is .....

(1) $\frac{2\pi}{3}$ (2) $\frac{3\pi}{4}$ (3) $\frac{5\pi}{4}$ (4) $\frac{\pi}{4}$

17. If the system of equation AX = B whose the last equivalent matrix is $\left(\begin{array}{ccc|c}1&2&3&1\\0&2&1&4\\0&0&\alpha+2&\beta-3\end{array}\right)$ has consistent and infinitely many solution when

(1) $\alpha=2$ and $\beta\neq3$ (2) $\alpha=-2$ and $\beta=3$ (3) $\alpha\neq-2$ and $\beta=3$ (4) $\alpha=2$ and $\beta=-3$

18. $[\hat{i}\times\hat{j},\ \hat{j}\times\hat{k},\ \hat{k}\times\hat{i}]=$

(1) 0 (2) 1 (3) 2 (4) 3

19. If $\omega\neq0$ is a fourth root of unity then the value of ω is

(1) $-\frac{1}{2}-i\frac{\sqrt{3}}{2}$ (2) $-\frac{1}{2}+i\frac{\sqrt{3}}{2}$ (3) i (4) $-i$

20. $sin^{-1}[sin4]=$

(1) 4 (2) $3\pi-4$ (3) $\pi-4$ (4) $2\pi-4$

PART - II

Note: 1. Answer any 7 Questions. 2. Q.No. 30 is Compoulsory. 7x2=14

21. Prove that $\begin{bmatrix}cos\theta & -sin\theta\\ sin\theta & cos\theta\end{bmatrix}$ is orthogonal.

22. Find the square root of $-6-8i$

23. Find the sum of the squares of the roots of the equation $2x^4-8x^3+6x^2-3=0$.

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24. Find the period and amplitude of $y = 4\sin(-2x)$

25. Prove that $[\vec{a} - \vec{b},\ \vec{b} - \vec{c},\ \vec{c} - \vec{a}] = 0$

26. Find the centre and radius of the circle $x^2 + y^2 + 6x - 4y + 4 = 0$.

27. Find the general equation of the circle whose diameter is the line segment joining the Points $(-4,-2)$ and $(1,1)$

28. If $2\hat{\imath} - \hat{\jmath} + 3\hat{k}$, $3\hat{\imath} + 2\hat{\jmath} + \hat{k}$, $\hat{\imath} + m\hat{\jmath} + 4\hat{k}$ are coplanar, find the value of m

29. Find the intercepts cut off by the plane $\vec{r} \cdot (3\hat{\imath} + 2\hat{\jmath} - 2\hat{k}) = 12$ on the co-ordinate axes

30. Find the rank of the matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 4 \\ 5 & 10 & -5 & 1 \end{bmatrix}$

PART - III

Note: 1. Answer any 7 Questions. 2. Q.No. 40 is Compoulsory. 7x3=21

31. Solve by Cramer's rule $5x - 2y + 16 = 0$, $x + 3y - 7 = 0$

32. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solution

33. Find the value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$

34. Find the vertex, focus, equation of directrix of the parabola $x^2 = -16y$

35. If $|z| = 2$ show that $3 \le |z + 3 + 4i| \le 7$

36. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$.

37. Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

38. Find the angle between the straight line $\vec{r} = (2\hat{\imath} + 3\hat{\jmath} + \hat{k}) + t(\hat{\imath} - \hat{\jmath} + \hat{k})$ and the plane $2x - y + z = 5$

39. A straight line passes through the point $(1,2,-3)$ and parallel to $4\hat{\imath} + 5\hat{\jmath} - 7\hat{k}$.
Find (i) vector equation in parametric form (ii) vector equation in non-parametric form
(iii) Cartesian equation of the straight line.

40. Find the equation of the in circle and circumcircle of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$.
Also find its eccentricity.

PART - IV

Note: 1. Answer all the Questions. 2. Each Questions Carries 5 Marks. 7x5=35

41. (a) If $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, Verify that $A(\text{adj}A) = (\text{adj}A)A = |A|I$

(OR)

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(b) Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates Of the point of contact.

42. (a) If $z = x + iy$ and $arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$.

(OR)

(b) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

43.(a) Find all zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1 + 2i$ and $\sqrt{3}$ are two of its zeros.

(OR)

(b) Prove by vector method $\sin(\alpha + \beta) = sin\alpha cos\beta + cos\alpha sin\beta$

44.(a) Solve $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$, if $6x^2 < 1$.

(OR)

(b) Find the equation of the circle through the points $(1,0)$, $(-1,0)$ and $(0,1)$.

45.(a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum of 4m when it is 6m away from the point of projection. Finally it reaches the ground 12m away from the starting point. Find the angle of projection?

(OR)

(b) Find the value of k for which the equations $kx - 2y + z = 1$, $x - 2ky + z = -2$, $x - 2y + kz = 1$ have (i) no solution (ii) unique solution (iii) infinitely many solution

46.(a) Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point $(1, -2, 4)$ and perpendicular to the plane $x + 2y - 3z = 11$ and parallel to the line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$.

(OR)

(b) If $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$, then show that

(i) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha + \beta + \gamma)$

(ii) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha + \beta + \gamma)$

47.(a) If $\vec{a} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}$, $\vec{b} = 3\hat{\imath} + 5\hat{\jmath} = 2\hat{k}$, $\vec{c} = -\hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ verify $(\vec{a} \times \vec{c}) \times \vec{b} = (\vec{a} \cdot \vec{b})\vec{c} - (\vec{c} \cdot \vec{b})\vec{a}$

(OR)

(b) Find the vector parametric, vector non-parametric and Cartesian form of the equation of the plane passing through the points (-1,2,0), (2,2, -1) and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.

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